

BAULKHAM HILLS HIGH SCHOOL



MATHEMATICS EXTENSION 1 ASSESSMENT

December 2012

*Time allowed: 50 minutes
plus 5 minutes reading time*

STUDENT NUMBER : _____

TEACHER'S NAME: _____



Extension 1 Mathematics

December 2012

Time: 50 minutes + 5 minutes reading time

DIRECTIONS

- Full working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Use black or blue pen only (*not pencils*) to write your solutions.
- No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.
- Approved Maths aids and calculators may be used

Question 1 (8 marks) – Start a new page

- a) For each of the following expressions, state whether or not they are polynomials. If it is a polynomial, state its degree, if it isn't explain why it isn't.

(i) $3^{-1}x - 4x^8 + \sqrt{3}$

1

(ii) $4x^7 + \sqrt{x}$

1

- b) If α, β and γ are the roots of $2x^3 - 6x^2 - 4x + 1 = 0$ find

(i) $\alpha + \beta + \gamma$

1

(ii) $\alpha\beta + \beta\gamma + \alpha\gamma$

1

(iii) $\alpha\beta\gamma$

1

(iv) $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma}$

1

(v) $\alpha^2 + \beta^2 + \gamma^2$

2

Question 2 (9 marks) – Start a new page

- a) (i) Sketch without the use of calculus, the polynomial $P(x) = (1 - 2x)^2(x + 1)^3$ showing x and y intercepts.

2

- (ii) Hence, or otherwise solve the inequation $P(x) > 0$

2

- b) A polynomial is given by $P(x) = x^3 + ax^2 + x - 18$

2

Find the values for a if -24 is the remainder when $P(x)$ is divided by $(x - 2)$

- c) For all positive integers of n , prove by mathematical induction that

3

$$\sum_{r=1}^n (r-1)^2 = \frac{n(n-1)(2n-1)}{6}$$

Question 3 (8 marks) – Start a new page

- a) Write a polynomial with degree 3, which is odd, has a zero of 2 and has a remainder 5 when divided by $x + 1$

3

- b) The roots of the equation $4x^3 + 6x^2 + c = 0$ are α, β and $\alpha\beta$ (where c is a non-zero constant)

- (i) Show that $\alpha\beta \neq 0$

1

- (ii) Show that $\alpha\beta + \alpha^2\beta + \alpha\beta^2 = 0$ and hence show that the value of $\alpha + \beta$ is -1

2

- (iii) Find the value of $\alpha\beta$

2

Question 4 (6 marks) – Start a new page		
a)	Use mathematical induction to show that the expression $7^n + 5$ is divisible by 6 for all positive integers n .	3
b)	Solve $4x^3 - 12x^2 + 11x - 3 = 0$ given that the roots are consecutive terms of an arithmetic series.	3
Question 5 (9 marks) – Start a new page		
a)	Given point P is a variable point with coordinates $(2p^2 + 1, 4)$	
	(i) Find the equation of the locus of P .	1
	(ii) State any restrictions of the point P .	1
b)	Two points $P(2p, p^2)$ and $Q(2q, q^2)$ lie on the parabola $x^2 = 4y$, and the line joining PQ is parallel to the line $y = mx$.	
	(i) Show that $p + q = 2m$	1
	(ii) Show that the equation of the normal to the parabola at the point P is $x + py = 2p + p^3$	2
	(iii) Show that the point of intersection of the normals from P and Q is $N(-pq(p + q), 2 + p^2 + pq + q^2)$	2
	(iv) Find the equation in Cartesian form of the locus of the point N .	2
~ End of Exam ~		

2012

Q1

- a) i) Yes - Degree 8
ii) No - $\sqrt{x} = x^{\frac{1}{2}}$
Polynomials must have powers with positive integers.

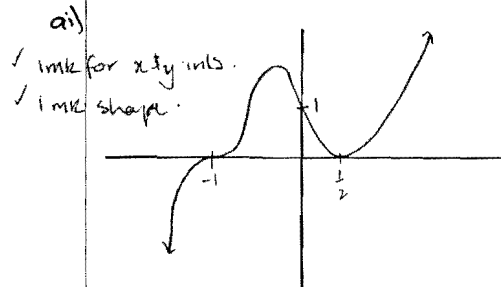
b) $2x^3 - 6x^2 - 4x + 1 = 0$

- i) $\frac{6}{2} = 3$
ii) $\frac{-4}{2} = -2$
iii) $-\frac{1}{2}$

iv) $\frac{\alpha + \beta + \gamma}{\alpha\beta\gamma} = \frac{3}{(-\frac{1}{2})}$

v) $(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \checkmark$
 $= (-2)^2 - 2(3)$
 $= 13$

Q2



ii) $x > -1$ and $x \neq \frac{1}{2}$ $\checkmark$

b) $-24 = (2)^3 + a(2)^2 + (2) - 19 \checkmark$
 $-16 = 4a$
 $a = -4$ $\checkmark$

c) $\sum_{r=1}^n (r-1)^2 = \frac{n(n-1)(2n-1)}{6}$

ie $0^2 + 1^2 + 2^2 + \dots + (n-1)^2 = \frac{n(n-1)(2n-1)}{6}$

For $n=1$

LHS = $0^2 = 0$

RHS = $1(1-1)(2-1) = 0$

LHS = RHS

$\therefore$ true for $n=1$ $\checkmark$

Assume true for $n=k$

ie $0^2 + 1^2 + 2^2 + \dots + (k-1)^2 = \frac{k(k-1)(2k-1)}{6}$

Prove true for $n=k+1$

Aim: $0^2 + 1^2 + 2^2 + \dots + (k-1)^2 + k^2$
 $= \frac{k(k+1)(2k+1)}{6}$

Proof:

LHS = $0^2 + 1^2 + 2^2 + \dots + (k-1)^2 + k^2$
 $= \frac{k(k-1)(2k-1)}{6} + k^2$ (by assumption)

$= \frac{k}{6} [(k-1)(2k-1) + 6k]$

$= \frac{k}{6} (2k^2 - 3k + 1 + 6k)$

$= \frac{k}{6} (2k^2 + 3k + 1)$

$= \frac{k}{6} (k+1)(2k+1)$

$=$ RHS.

$\therefore$ If it is true for $n=k$ & proven true for $n=k+1$ & since proven true for $n=1$ then it is true for $n=2, 3, 4, \dots$ ie all positive integers of n

Note: $P(x) = a(x)(x+2)(x-2)$
 $P(-1) = 5 = a(-1)(-1+2)(-1-2)$
 $a = \frac{5}{3}$

Q3

a) $P(x) = ax^3 + bx^2 + cx + d$
 $P(-x) = -ax^3 + bx^2 - cx + d$

Since $P(x)$ is odd

ie $P(-x) = -P(x)$

$P(-x) + P(x) = 0$

then $b = d = 0$

$\therefore P(x) = ax^3 + cx$ $\checkmark$

$P(2) = 0 = 8a + 2c$

$0 = 4a + c$ — (1) $\checkmark$

$P(-1) = 5 = -a - c$ — (2) $\checkmark$

(1) + (2) $5 = 3a$

$a = \frac{5}{3}$

(2) $\rightarrow 5 = -\frac{5}{3} - c$

$c = -\frac{20}{3}$

$\checkmark$ Imk correct

$\therefore P(x) = \frac{5}{3}x^3 - \frac{20}{3}x$ form from Simult. Solns.

b) $4x^3 + 6x^2 + 0x + c = 0$

i) $\alpha\beta(\alpha\beta) = -\frac{c}{4}$

$\left\{ \begin{array}{l} \alpha^2\beta^2 = -\frac{c}{4} \\ \alpha\beta = \pm \sqrt{-\frac{c}{4}} \text{ but } c \neq 0 \\ \therefore \alpha\beta \neq 0 \end{array} \right.$

ii) $\alpha\beta + \alpha(\alpha\beta) + \beta(\alpha\beta) = \frac{0}{4}$ $\checkmark$

$\alpha\beta(1 + \alpha + \beta) = 0$

$\alpha\beta \neq 0$ (part i) $1 + \alpha + \beta = 0$
 $\alpha + \beta = -1$ $\checkmark$

ii) $\alpha + \beta + (\alpha\beta) = \frac{-6}{4}$ $\checkmark$

$-1 + \alpha\beta = -\frac{3}{2}$

$\alpha\beta = -\frac{1}{2}$ $\checkmark$

Q4

a) For $n=1$

$7^1 + 5 = 12 = 6 \times 2$, divisible by 6

$\therefore$ true for $n=1$ $\checkmark$

Assume true for $n=k$

$7^k + 5 = 6P$

where P is an integer

Investigate for $n=k+1$

Aim: $7^{k+1} + 5 = 6Q$ where Q is an integer.

Proof: LHS = $7^{k+1} + 5$
 $= 7 \times 7^k + 5$
 $= 7(6P - 5) + 5$ by assume
 $= 7 \times 6P - 30$
 $= 6(7P - 5)$ since $(7P - 5)$ is an integer.
 $= 6Q$

$\therefore$ divisible by 6 for $n=k+1$

if divisible by 6 for $n=k$

$\therefore$ If it is true for $n=k$, and true for $n=k+1$ and since it is true for $n=1$, then it is true for $n=2, 3, \dots$ ie all positive integers of n .

Q4 cont.

b) $4x^3 - 12x^2 + 11x - 3 = 0$

Let the roots be $\alpha-d, \alpha, \alpha+d$

$$(\alpha-d) + \alpha + (\alpha+d) = \frac{12}{4}$$

$$3\alpha = 3$$

$$\alpha = 1$$

$$\alpha(\alpha-d)(\alpha+d) = -\frac{-3}{4}$$

but $\alpha=1 \dots \rightarrow (1-d)(1+d) = \frac{3}{4}$

$$1-d^2 = \frac{3}{4}$$

$$d^2 = \frac{1}{4}$$

$$d = \pm \frac{1}{2}$$

$\therefore$ roots are $1-\frac{1}{2}, 1, 1+\frac{1}{2}$

$$\therefore x = \frac{1}{2}, x = 1, x = \frac{3}{2}$$

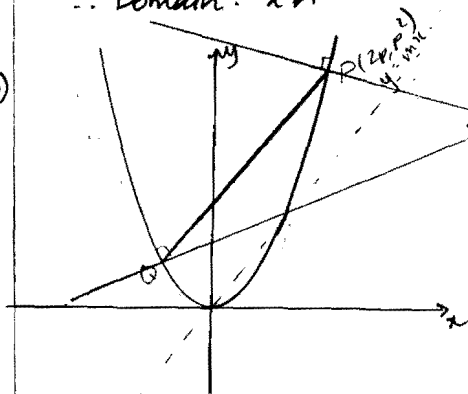
Q5

a) i) $y=4$

ii) Since $x=2p^2+1$

$$\therefore \text{Domain: } x \geq 1$$

b)



i) $m_{PQ} = \frac{p^2 - q^2}{2(p+q)}$

$$-n = \frac{(p+q)(p-q)}{2(p+q)}$$

$$2m = p+q \quad \checkmark$$

ii) $y = \frac{x^2}{4}$

$$\frac{dy}{dx} = \frac{x}{2}$$

$$m_T = \frac{2p}{2} = p$$

$$\therefore m_N = -\frac{1}{p} \quad \checkmark$$

Eqn of normal

$$y - p^2 = -\frac{1}{p}(x - 2p)$$

$$-yp + p^3 = x - 2p \quad \checkmark$$

$$x + yp = p^3 + 2p \quad \text{--- (1)}$$

ii) Normal at Q is $x + qy = 2q + q^3$ --- (2)

$$\textcircled{1} - \textcircled{2} \quad (p-q)y = 2(p-q) + (p^3 - q^3) \quad \checkmark$$

$$y = 2 + p^2 + pq + q^2$$

Sub $y \rightarrow \textcircled{2}$

$$x + p(2 + p^2 + pq + q^2) = p^3 + 2p$$

$$x + 2p + p^3 + p^2q + pq^2 = p^3 + 2p$$

$$x = -p^2q - pq^2$$

$$x = -pq(p+q)$$

$$\therefore N(-pq(p+q), p^2 + pq + q^2 + 2)$$

Q5 cont.

iv) $x = -pq(p+q)$

but $p+q = 2m$ $\leftarrow$ using this

$$x = -pq(2m)$$

$$pq = \frac{-x}{2m}$$

$$y = p^2 + pq + q^2 + 2 + p^2 - pq$$

$$y = (p+q)^2 + 2 - pq$$

but $p+q = 2m$
 $\& \quad pq = \frac{-x}{2m} \quad \checkmark$

$$y = (2m)^2 + 2 - \left(\frac{-x}{2m}\right)$$

$$y = 4m^2 + 2 + \frac{x}{2m} \quad \checkmark$$